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Lab Week 8

Challenge 1 : NetBIOS Enumeration

```
(kalikali@kali)-[~]
$ nmblookup -A 192.168.56.103
Looking up status of 192.168.56.103
    METASPLOITABLE <00> - B <ACTIVE>
    METASPLOITABLE <03> - B <ACTIVE>
    METASPLOITABLE <20> - B <ACTIVE>
    ._.MSBROWSE_. <01> - <GROUP> B <ACTIVE>
    WORKGROUP <00> - <GROUP> B <ACTIVE>
    WORKGROUP <1d> - B <ACTIVE>
    WORKGROUP <1e> - <GROUP> B <ACTIVE>

    MAC Address = 00-00-00-00-00-00
```

Challenge 2 : Fast Nmap Scan

```
(kalikali@kali)-[~]
$ nmap 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-06 15:13 -0400
Nmap scan report for 192.168.56.103
Host is up (0.000062s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
```

```
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:5D:78:A4 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.14 seconds
```

Challenge 3 : DNS Records

```
(kalikali@kali)-[~]
$ nslookup
> example.com
Server:      192.168.0.1
Address:     192.168.0.1#53

Non-authoritative answer:
Name:   example.com
Address: 104.20.23.154
Name:   example.com
Address: 172.66.147.243
Name:   example.com
Address: 2606:4700:10::ac42:93f3
Name:   example.com
Address: 2606:4700:10::6814:179a
>
```

```
(kalikali@kali)-[~]
$ dig google.com

; <<>> DiG 9.20.20-1-Debian <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 1856
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;google.com.                IN      A
;; ANSWER SECTION:
google.com.                 59      IN      A      172.217.25.142

;; Query time: 1100 msec
;; SERVER: 192.168.0.1#53(192.168.0.1) (UDP)
;; WHEN: Sat Jun 06 15:24:07 EDT 2026
;; MSG SIZE rcvd: 55
```

```

(kalikali@kali)-[~]
$ dig mx yahoo.com

; <<>> DiG 9.20.20-1-Debian <<>> mx yahoo.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 12984
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags::; udp: 512
;; QUESTION SECTION:
;yahoo.com.                IN      MX

;; ANSWER SECTION:
yahoo.com.                900     IN      MX      1 mta5.am0.yahoodns.net.
yahoo.com.                900     IN      MX      1 mta7.am0.yahoodns.net.
yahoo.com.                900     IN      MX      1 mta6.am0.yahoodns.net.

;; Query time: 192 msec
;; SERVER: 192.168.0.1#53(192.168.0.1) (UDP)
;; WHEN: Sat Jun 06 15:24:56 EDT 2026
;; MSG SIZE rcvd: 117

```

Challenge 4 : TTL OS Fingerprinting

```

(kalikali@kali)-[~]
$ ping 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data.

```

Challenge 5 : Anonymous LDAP Query

```

(kalikali@kali)-[~]
$ ldapsearch -x -H ldap://192.168.56.103 -b "dc=example,dc=com"
ldap_sasl_bind(SIMPLE): Can't contact LDAP server (-1)

```

Challenge 6 : Anonymous FTP Login

```
(kalikali@kali)-[~]
$ ftp 192.168.56.103
Connected to 192.168.56.103.
220 (vsFTPD 2.3.4)
Name (192.168.56.103:kalikali):
331 Please specify the password.
Password:

530 Login incorrect.
ftp: Login failed
ftp>
ftp> ls -a
530 Please login with USER and PASS.
530 Please login with USER and PASS.
ftp: Can't bind for data connection: Address already in use
ftp>
```

Challenge 7 : SMB NSE Enumeration

```
(kalikali@kali)-[~]
$ nmap --script smb-os-discovery -p445 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-06 15:47 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.
Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.103
Host is up (0.00029s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:5D:78:A4 (Oracle VirtualBox virtual NIC)

Host script results:
| smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_  System time: 2026-06-06T15:47:24-04:00

Nmap done: 1 IP address (1 host up) scanned in 0.35 seconds
```

```

(kalikali@kali)-[~]
└─$ nmap --script smb-enum-users -p445 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-06 15:47 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.
Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.103
Host is up (0.00036s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:5D:78:A4 (Oracle VirtualBox virtual NIC)

Host script results:
| smb-enum-users:
|   METASPLOITABLE\backup (RID: 1068)
|     Full name: backup
|     Flags: Account disabled, Normal user account
|   METASPLOITABLE\bin (RID: 1004)
|     Full name: bin
|     Flags: Account disabled, Normal user account
|   METASPLOITABLE\bind (RID: 1210)
|     Flags: Account disabled, Normal user account
|   METASPLOITABLE\daemon (RID: 1002)
|     Full name: daemon

```

Challenge 8 : Enum4linux

```

(kalikali@kali)-[~]
└─$ enum4linux -a 192.168.56.103
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux
/ ) on Sat Jun 6 15:49:58 2026

===== ( Target Information ) =====
=====
Target ..... 192.168.56.103
RID Range ..... 500-550,1000-1050
Username ..... ''
Password ..... ''
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none

===== ( Enumerating Workgroup/Domain on 192.168.56.103 ) =====
=====

[+] Got domain/workgroup name: WORKGROUP

===== ( Nbtstat Information for 192.168.56.103 ) =====
=====

```

Challenge 9 : SNMP NSE

```
(kalikali@kali)-[~]  
$ nmap -sU -p161 --script=snmp-processes 192.168.56.103  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-06 15:52 -0400  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.  
Try using --system-dns or specify valid servers with --dns-servers  
Nmap scan report for 192.168.56.103  
Host is up (0.00036s latency).  
  
PORT      STATE SERVICE  
161/udp   closed snmp  
MAC Address: 08:00:27:5D:78:A4 (Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.39 seconds
```

```
(kalikali@kali)-[~]  
$ nmap -sU -p161 --script=snmp-sysdescr 192.168.56.103  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-06 15:52 -0400  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.  
Try using --system-dns or specify valid servers with --dns-servers  
Nmap scan report for 192.168.56.103  
Host is up (0.00056s latency).  
  
PORT      STATE SERVICE  
161/udp   closed snmp  
MAC Address: 08:00:27:5D:78:A4 (Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.40 seconds
```


Challenge 10 : OS Detection

```
(kalikali@kali)-[~]  
$ nmap -O 192.168.56.103  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-06 15:53 -0400  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.  
Try using --system-dns or specify valid servers with --dns-servers  
Nmap scan report for 192.168.56.103  
Host is up (0.00039s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
8009/tcp  open  ajp13  
8180/tcp  open  unknown  
MAC Address: 08:00:27:5D:78:A4 (Oracle VirtualBox virtual NIC)  
Device type: general purpose  
Running: Linux 2.6.X  
OS CPE: cpe:/o:linux:linux_kernel:2.6  
OS details: Linux 2.6.9 - 2.6.33  
Network Distance: 1 hop  
  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 1.64 seconds
```